Chapter 1: Introduction to Biology

Biology is the scientific study of living organisms and their interactions with the environment.

This handbook covers cells, energy, genetics and ecosystems across twelve chapters.

Chapter 2: Cell Theory

Cell theory states that all living things are made of cells and that the cell is the basic unit of life.

Robert Hooke first described cells in 1665 while observing cork under a microscope.

Chapter 3: The Cell Membrane

The cell membrane is a phospholipid bilayer that controls what enters and leaves the cell.

Membrane proteins act as channels, pumps and receptors for signaling molecules.

Chapter 4: Mitochondria

Mitochondria are the powerhouse of the cell and produce ATP through cellular respiration.

Each mitochondrion has its own circular DNA inherited from the mother.

Chapter 5: Photosynthesis

Photosynthesis converts light energy, water and carbon dioxide into glucose and oxygen inside chloroplasts.

The light reactions happen in the thylakoid membranes and the Calvin cycle happens in the stroma.

Chapter 6: DNA Structure

DNA is a double helix made of nucleotides; adenine pairs with thymine and guanine pairs with cytosine.

James Watson and Francis Crick described the double helix structure in 1953.

Chapter 7: Protein Synthesis

Protein synthesis has two stages: transcription in the nucleus and translation at the ribosome.

Messenger RNA carries the genetic code, and transfer RNA brings matching amino acids.

Chapter 8: Mendelian Genetics

Gregor Mendel studied pea plants and discovered dominant and recessive traits.

His law of segregation says each parent passes one allele for each trait to the offspring.

Chapter 9: Evolution

Natural selection is the process where organisms better adapted to their environment survive and reproduce more.

Charles Darwin published On the Origin of Species in 1859.

Chapter 10: Ecosystems

An ecosystem includes all living organisms in an area together with the non-living environment.

Energy flows from producers to primary consumers to secondary consumers, losing about ninety percent at each level.

Chapter 11: The Nitrogen Cycle

Nitrogen-fixing bacteria in the soil convert atmospheric nitrogen into ammonia that plants can use.

Denitrifying bacteria return nitrogen gas to the atmosphere.

Chapter 12: Summary

Living systems depend on cells, energy conversion, inherited information and interactions within ecosystems.

Review each chapter and test yourself with the practice questions.